711 Problem set 4

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```
rm(list=ls())
set.seed(123)
```

2. Problem to demonstrate the role of qualitative (nominal) predictors in addition to quantitative predictors in multiple linear regression.

Attach “Credits” data from R. Regress “balance” on: (a) “gender” only.

(b) “gender” and “ethnicity” .

(c) “gender”, “ethnicity”, “income”.

(d) Output all the regressions in (a)-(c) in a single table using stargazer. Comment on the significant coefficients in each of the models.

(e) Explain how gender affects “balance” in each of the models (a)- (c) .

(f) Compare the average credit card balance of a male African with a male Caucasian on the basis of model (b).

(g) Compare the average credit card balance of a male African with a male Caucasian when each earns 100,000 dollars. For comparison, use the model in (c).

(h) Compare and comment on the answers in (f) and (g)

(i) Based on the model in (c), predict the credit card balance of a female Asian whose income is 2000,000 dollars.

(j) Check the goodness of fit of the different models in (a) -(c) in terms of AIC, BIC and adjusted R2. Which model would you prefer?

Solution 2:

```
rm(list=ls())
library(ISLR)
attach(Credit)
??Credit

## starting httpd help server ... done

View(Credit)
```

(a) Regression of Balance on Gender

```r
m1=lm(Balance~Gender)
summary(m1)
```

```
##
## Call:
## lm(formula = Balance ~ Gender)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -529.54 -455.35  -60.17  334.71 1489.20
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    509.80      33.13  15.389   <2e-16 ***
## GenderFemale    19.73      46.05   0.429    0.669
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 460.2 on 398 degrees of freedom
## Multiple R-squared:  0.0004611,  Adjusted R-squared:  -0.00205
## F-statistic: 0.1836 on 1 and 398 DF,  p-value: 0.6685
```

Predicted Model: $\widehat{Balance} = 509.8 + 19.73Gender_{Female}$

Interpretation: When the Gender is male, predicted balance is 509.8 units when the Gender is female, predicted balance is 529.53 units

(b) Regression of Balance on Gender and Ethnicity

```r
m2=lm(Balance~Gender+Ethnicity)
summary(m2)
```

```
##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -540.92 -453.61  -56.37  336.24 1490.77
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)          520.88      51.90  10.036   <2e-16 ***
## GenderFemale          20.04      46.18   0.434    0.665
## EthnicityAsian       -19.37      65.11  -0.298    0.766
## EthnicityCaucasian   -12.65      56.74  -0.223    0.824
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 461.3 on 396 degrees of freedom
```

```
## Multiple R-squared:  0.000694,   Adjusted R-squared:  -0.006877
## F-statistic: 0.09167 on 3 and 396 DF,  p-value: 0.9646
```

Predicted Model: $\widehat{Balance} = 520.88 + 20.04Gender_{Female} - 19.37Ethnicity_{Asian} - 12.65Ethnicity_{Caucasian}$

Interpretation: When Gender is Male and the Ethnicity is African, the Balance is 520.88 units

(c) Regression of Balance on Gender, Ethnicity, Income.

```
m3=lm(Balance~Gender+Ethnicity+Income)
summary(m3)

##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity + Income)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -794.14 -351.67  -52.02  328.02 1110.09
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         230.0291    53.8574   4.271 2.44e-05 ***
## GenderFemale         24.3396    40.9630   0.594    0.553
## EthnicityAsian        1.6372    57.7867   0.028    0.977
## EthnicityCaucasian    6.4469    50.3634   0.128    0.898
## Income                6.0542     0.5818  10.406  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 409.2 on 395 degrees of freedom
## Multiple R-squared:  0.2157, Adjusted R-squared:  0.2078
## F-statistic: 27.16 on 4 and 395 DF,  p-value: < 2.2e-16
```

Predicted Model: $\widehat{Balance} = 230.0291 + 24.3396Gender_{Female} + 1.6372Ethnicity_{Asian} + 6.4469Ethnicity_{Caucasian} + 6.0542Income$

(d) Comparing 3 models in Stargazer.

```
library(stargazer)

##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary
Statistics Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

stargazer(m1,m2,m3,type="text")
```

```
##
##
================================================================================
======
##                                            Dependent variable:
##                       -------------------------------------------------------
---------
##                                                  Balance
##                              (1)                 (2)                   (3)
## ---------------------------------------------------------------------------
---------
## GenderFemale               19.733              20.038                24.340
##                           (46.051)            (46.178)
(40.963)
##
## EthnicityAsian                                 -19.371                1.637
##                                               (65.107)
(57.787)
##
## EthnicityCaucasian                             -12.653                6.447
##                                               (56.740)
(50.363)
##
## Income
6.054***
##
(0.582)
##
## Constant                  509.803***          520.880***
230.029***
##                           (33.128)            (51.901)
(53.857)
##
## ---------------------------------------------------------------------------
---------
## Observations                 400                 400                   400
## R2                         0.0005               0.001                 0.216
## Adjusted R2                -0.002              -0.007                 0.208
## Residual Std. Error 460.230 (df = 398)  461.337 (df = 396)    409.218 (df
= 395)
## F Statistic         0.184 (df = 1; 398) 0.092 (df = 3; 396) 27.161*** (df
= 4; 395)
##
================================================================================
======
## Note:                                                  *p<0.1; **p<0.05;
***p<0.01
```

(e) We see that as other factors are added the effect of gender is increased. As more factors are considered, magnitude of increase in balance due to the gender being female increases.

(f) We know the difference will be the the coefficient for Ethnicity: Caucasian.

(g) Compare the average credit card balance of a male African with a male Caucasian on the basis of model (b).

```
balance_m_afa=230.029 + 6.054*100
balance_m_afa

## [1] 835.429

balance_m_cauc= 230.029 + 6.054*100 + 6.447
balance_m_cauc

## [1] 841.876

#difference is the coefficient for Ethnicity Caucasian
```

We see that the difference is 6.447

(h) Compare and comment on the answers in (f) and (g): We see that the difference is same in both cases. We can infer that the income being 0 or being fixed at a particular level the output remains unchanged.

(i) Based on the model in (c), predict the credit card balance of a female Asian whose income is 2000,000 dollars.

```
230.0291 +  24.3396 + 1.6372 + 6.0542*2000

## [1] 12364.41
```

the predicted balance is $12364410.

(j) Check the goodness of fit of the different models in (a) -(c) in terms of AIC, BIC and adjusted R2. Which model would you prefer?

```
model = c(paste("model:",c("(a)","(b)","(c)")))
R_sq = c(summary(m1)$r.squared,summary(m2)$r.squared,summary(m3)$r.squared)

data.frame(model,R_sq)

##         model        R_sq
## 1 model: (a) 0.000461133
## 2 model: (b) 0.000693986
## 3 model: (c) 0.215716091
```

We would prefer model (c) as it explains 21% of the total variation in Balance.

4. Problem to demonstrate the impact of ignoring interaction term in multiple linear regression

Consider a simulation setting where the data is generated as follows:

Step 1: Generate x_{1i} from Normal(0,1) distribution, i = 1, 2, .., n

Step 2: Generate x_{2i} from Bernoulli(0.3) distribution, i = 1, 2, .., n

Step 3: Generate ε_i from Normal(0,1) and hence generate the response:

$yi = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 (x_{1i}.x_{2i}) + \varepsilon_i$, i = 1, 2, , , n.

Step 4: Run two separate multiple linear regressions (i) using the model in Step 3 and (ii) using the model in Step 3 without the interaction term.

Repeat Steps 1-4 , R = 1000 times. At each simulation compute the MSE for the correct model (i.e. model with the interaction term) and the naive model (i.e. the model without the interaction term). Finally find the average MSE's for each model. From the output, demonstrate the impact of ignoring the iteraction term.)

Carry out the analysis for n = 100 and the following parametric configurations: $(\beta_0, \beta_1, \beta_2, \beta_3) = (-2.5,1.2,2.3,0.001), (-2.5,1.2.2.3,3.1)$. Set seed as 123.

Solution 4.

```
compare_MSE = function(beta,n,R)
{
  mse_correct=c()
  mse_naive=c()
  for(i in 1:R)
  {
    x1 = rnorm(n)
    x2 = rbinom(n,1,0.3)
    e = rnorm(n)

    y = beta[1]+beta[2]*x1+beta[3]*x2+beta[4]*(x1*x2)+e

    model1 = lm(y~ x1 + x2+ I(x1*x2))
    mse_correct[i] =  mean((y-predict(model1))^2)

    model2 = lm(y~x1+x2)
    mse_naive[i] = mean((y-predict(model2))^2)
  }
  cat("Average Correct MSE:", mean(mse_correct),
      "\tAverage Naive MSE:", mean(mse_naive), "\n\n")
}
```

```r
beta1=c(-2.5,1.2,2.3,0.001)
compare_MSE(beta1,100,1000)

## Average Correct MSE: 0.9606393   Average Naive MSE: 0.9703833

beta2=c(-2.5,1.2,2.3,3.1)
compare_MSE(beta2,100,1000)

## Average Correct MSE: 0.9575532   Average Naive MSE: 2.87125
```